Data and Data Cleaning

Howles

Data

- We'll look at:
  - How it is represented
  - What is dirty data?
  - Conversions that may be necessary
  - Missing or inaccurate data
  - How we will prepare "scrub" the data
  - Other data management issues

CRISP-DM

- In the CRISP-DM model, steps involving the data are the most dominate and important
- Data preparation and cleaning accounts for most of the data mining effort (on average)

Data-Related Steps in CRISP-DM

- Select/find the data
- Clean the data
- Construct data
- Integrate data
- Format data

Data Terms

- Attributes, fields, variables, features, indicators, columns
- Instances, records, observations, rows, cases, objects
- Data set

How Data is Represented

- Numeric – continuous attributes. Measurements, int, float data types
- Nominal – values are symbolic - labels: sunny, old, yellow,; can perform equality checks. Categorical coding; may be code “1” but has no arithmetic meaning
- Ratio – the measurement scheme defines a zero point (a distance, temperature differential but not the temperature reading itself because it does not inherently define a zero point); math operations are valid

How Data is Represented

- Ordinal – rank order: “cold, cool, warm, hot” or “good, better, best”. No “distance” between them; can perform equality checks
  - No distance measure does not matter so long as we’re consistent

How Data is Represented

- Interval – ordered and measured in fixed units: 1939 – 1946 .. But summing them up makes no sense
  - Units are fixed and equal
  - Example: temperature readings
- Statisticians use the same data typing but may have slightly different names

Data

- Numeric data often easy to interpret if defined ranges exist (>32 & water, not frozen; 1 – 12 months…)
- How do you measure something good, bad, healthy?
- Need a domain expert

Cautions about Cleaning

- Document what you do
- Work carefully
- Don’t make assumptions – work closely with your domain expert
- Be aware of bias

Introducing Bias

- Language – different terms or grammars to describe the domain, data attributes or the problem
- Search – look at other search options
- Overfitting – results provide a solution based on bad assumptions/patterns, or search stops too soon

Introducing Bias

– Actions already performed on the data
– How the data was gathered
  - How questions were asked
  - How responses were interpreted
  - Who asked the questions
  - How samples were selected

– In DM, often a synonym for “error”

Data Cleaning (Scrubbing)

- An important aspect of DM
- Handling invalid values, duplicates, missing data, data entry errors
- If necessary, converting data to specific values in order to perform correct measurements (putting a value to good, bad, healthy …)

Identifying Dirty Data

This process may appear to be trivial, but we are working with very large data sets

Suppose we have thousands or millions of records?

What Do We Mean by Dirty Data

- Data that is incorrect, inaccurate, irrelevant or incomplete
- Data needing to be converted (nominal to numeric)
- Data with different formats or coding schemes (dates are a common problem; more later)
- Data from > 1 file with different field delimiters
- Data that is coded
- Data that must be summarized “rolled up”

How does data get “dirty”?

- Inconsistent definitions, meanings (especially when combining different sources)
- Data entry mistakes
- Collection errors
- Corrupted data transmissions
- Conversion errors

Data Issues

- Out of range entries
- Unknown, unrecorded or irrelevant data
- Missing values (more later)
- Language translation issues

Data Issues

- Unavailable readings
- Inapplicable data (asking a male if pregnant)
- Customer provided incorrect data (POS zip codes)
- Duplicate data
  - Mailing vs billing addresses
  - Duplicate account numbers

Data Issues

- Stale data – old address or phone #
- Unavailable data – initially purchased from a data broker but no longer available
- Data may be available but it’s not in electronic form

Data Issues

- Data associated with the wrong person
  – Customer forgot frequent shopper card so the cashier used his/her own
- User provided wrong data
  – An online form forced a response; user entered garbage

Perspectives

- Data about the same entity may look very different, depending on the owner and use
- Sales, billing departments may be interested in different ways about the same data

Conversions

- Data may need to be mapped to new values
- Converted to other units
- Converted to other data types

Nominal to Numeric Conversions

- Some DM tools can internally convert
- Some tools may require only numeric inputs
- Some tools perform internal conversions on the fly, and (perhaps) without your knowledge/consent ← Beware!

Other conversions

- How would you convert binary fields to numeric values? Gender: M/F
- How would you convert ordered attributes to numeric values? Grades: A B C D F
  - Be consistent
  - Preserve natural order so comparison are valid

Other conversions

- How would you convert ordered attributes to numeric values? Grades: A B C D F
  - Be consistent
  - Preserve natural order so comparison are valid
  - What if +/- grades are entered?

Other Conversions

- Suppose you have info on the 50 United States
  - Establish 50 codes
  - Group into X regions
- How about 7000 professional codes but only a few are used?
  - Select most frequent
  - Group the rest
  - Watch your tool – some will convert to 7000 columns with sparse data

Dates

- Transform dates to a common format
  - 9/24/09, 24.09.09, Sept. 24, 2009, 24 Sept. 09
- Sometimes, just the year (yyyy) may be sufficient
- For more details, we may need the month, day, hour, second …

Dates

- How about this:
- Representing dates as YYYYMM or YYYYMMDD
  - What's good about this formatting?
  - What is the limitation?

Dates

- How about this:
- Representing dates as YYYYMM or YYYYMMDD
- What's good about this formatting?
- What is the limitation?
  - You can sort the data
  - Does not preserve intervals

Dates

- YYYYMMDD does not preserve intervals
  - Example: 20040201 – 20040131
  - This is 2 days

Date Options

- To preserve intervals, we can use
  - Unix system date: Number of seconds since 1970
  - Number of days since Jan 1, 1960 (SAS)
  - Use the Julian date
- Problems
  - Values are not obvious
  - Not intuitive
  - Harder to verify, easy to make mistakes

Legacy Issues

- Y2K – 2 digit year
  - Year 02 – is it 1902 or 2002?
- Depends on context (child's birthday or year a house was built)
- Typical approach: Set a cutoff year. If YY < cutoff, then 20YY else 19YY

Dealing with Missing Data

- Values may be missing due to
  - Unknown, unrecorded, irrelevant data
  - Malfunctioning equipment
  - Changes in the design
  - Collation/merge of different datasets
  - Unavailable data

Dealing with Missing Data

- Values may be missing due to
  - Removals because of security or privacy issues
  - Translation issues (especially languages)
  - The data being used for a different purpose than originally planned
  - Self-reporting – people may omit if the input mechanism does not require an input

What to do

- It may not matter, or it may matter a lot (consider missing test results when making a medical diagnosis)
- Action may depend on what you are doing, the project itself (goals), or the DM algorithm or tool you are using

Dealing with Missing Values

- Ignore the attribute or entire instance
  - Not desired because this may have a big impact on the N for your problem
- Try to estimate or predict; use mean, mode or median values
  - This approach is relatively easy and not bad, on average
- Treat as a separate value (male, female, missing)

Dealing with Missing Values

- Look for <empty>, zeroes, “.”, 999, N/A (among others); Decide on a standard; create a new value

Dealing with Missing Values

- Does missing imply a default value? Example: in a donation DB, an empty cell may mean the donor responded but did not send $$, or the donor didn't respond. Either results in a zero donation but for reasons that may be of interest to you
- If missing, leave blank or "missing" – this may mean something (e.g. fraud)

Dealing with Missing Values

- A specific example (Abbott):
  - A donation database with age as an attribute
  - You know that people tend to contribute more $$ if they are older
  - Lots of ages are zero
    - Does this represent a donation in honor of a newborn or missing data?
  - In this case, the missing data results in a non-working model

Dealing with Missing Values

- Impute the value based on previous values
  - Value was 100 in Q1, Q2 and Q4. Q3 is missing. High likelyhood Q3 is also 100

Dealing with Missing Values

- Suppose you were missing a number of data values and inserted zeros
- What have you now done to the mean and standard deviation?
- Think about what you're doing and how it may impact other operations
- Also important if using a DM algorithm relying on distances, means or standard deviations

Dealing with Missing Values

- Be careful when using tools:
  - Some tools have a default operation to handle missing data (add zeroes, use the mean, impute).

Dealing with Missing Values

- Be careful when using tools:
  - Some tools will randomly select values from the current distribution
    - Consider salary info with missing values. The tool will build a distribution of salaries then randomly pick
  - Advantage is that adding values randomly selected from a normal distribution to an existing normal distribution won't change the overall shape of the curve – little impact on the mean

Feature Creation

- A feature is a new version of an attribute derived from existing attributes
- Example:
  - You have customer age and know it will speed up your model if you add a feature "Age > 65"
- Separate if variables interact or combine if it can simplify

Feature Creation

- Here's another case (Abbott):
  - Government was researching fraudulent invoices. People were submitting invoices and slightly modifying name and mailing address
  - Initially slid under the radar of the program looking for multiple payments to the same person/address
  - Started stripping out special symbols in name/ address and storing as separate attribute

Problem

- You want your model to consider historical data, but historical data is not in the current data set

Dealing with Missing Values

- Again, missing values may not matter (if the field is not highly correlated to your goals) or it may matter a lot (some algorithms such as decision trees don't work well with missing values)
- It's better to not guess. Work with your domain expert for guidance on how to proceed

Inaccurate Data

- Data entry mistakes
- Measurement errors
- Outliers previously removed
- Duplicates
- Stale data – closed accounts, outdated phone numbers
- Different values, New York, NY, N.Y.

Finding Inaccurate Data

- Look for the obvious
  - Run statistical tools
  - Not always a problem, but may indicate an inconsistency
- Look for nonsensical data
  - Negative grade or age

Discretization

- Binning
- Useful for generating summary data
- Produces discrete values

Equal-Width

- Take the following 14 temperature readings:
- 64 65 68 69 70 71 72 72 75 75 80 81 83 84
- Group into bins of range size = 3
- [64-67), [67-70), [70-73), [73-76), [76-79), [79-82), [82-85)
- [= inclusive;) = exclusive
- Equal width, low <= value < high

Equal-Width

- Using equal-width with the previous data yields bins as shown:
- 64 65 68 69 70 71 72 72 75 75 80 81 83 84
- 2 [64-67), 2 [67-70), 4 [70-73), 2 [73-76), 0 [76-79), 2 [79-82), 2 [82-85)

Equal-width may result in clumping

- Consider salary ranges
  - 99% of all employees may earn between 0-200,000. The owner makes 2,000,000.
  - With a width of 200,000, only one person in the upper bin
- How can we even out the distribution?

Discretization: Equal-height

- Instead of defining the bin ranges of size N, assign N values to each bin. Here N = 3
- The last bin may have an != height
- 64 65 68 69 70 71 72 72 75 75 80 81 83 84
- What happened?

Discretization

- Equal-height preferred because it avoids clumping
- Equal-width is simplest; good for many situations
- In practice, “almost-equal” height binning is used; avoids clumping and gives more intuitive breakpoints

Discretization

- When binning:
  - Do not split repeated values across bins
  - Create a separate bin for special values (for example, the previous salary example)
- Other methods exist

Considerations

- Equal-width is simplest but not good for unequal distributions
- Equal-height gives better results
- Is it needed? Depends on what you are doing

Binning

- After you create bins, create a histogram of values
- Look at general shape of the histogram
- Jagged shape may indicate a weakness in the way the bins were formed
- Try different number of bins and different boundaries (shift ranges)

Binning

- Be careful – you can introduce unintended artifacts if bins are not carefully constructed
- Think about your data and the desired outcomes – if you are using salary, you may be able to bin the data into just a few bins, making analysis easier and (possibly) reducing your run time

Rollup

- Can help reduce the complexity of your model
  – You have 50 different attributes for state code; rollup to one attribute with specific values
  – For “wide” data values, can you break into regions or otherwise associated “buckets”?

Rollup

- Good idea anyway because lots of attributes with small counts are ‘usually’ insufficient in the models
- Also, the appearance of 50 columns may appear to be influential
- Also see similar problems with zip codes

Speaking of zip codes:

- How many zip codes exist?
- Does zip code matter in modeling?
  - For example in mortgage lending?

Speaking of zip codes:

- How many zip codes exist?
- Does zip code matter in modeling?
  - For example in mortgage lending?
- Some legal issues here – Red Lining
- May be considered discriminatory

“Outliers”

- In class, I use the term loosely
- My definition means “any value that doesn’t really look like most of the others in the data set”
- This means an “outlier” may be just a unique data point, or it really could be an outlier (an error, noise)

Outliers

- Outliers are data instances with characteristics that are considerably different than most of the other data in the set.
  - Values out of range
- What to do?
  - Nothing
  - Enforce upper and lower bounds
  - Let binning handle the problem

Outliers

- With a normal distribution, they could be > 3 standard deviations from your mean
- With a bimodal distribution, they could be in the middle or at the ends

Outliers

- Some are easy to find:
  - Negative age or age > 120
  - Negative number of children
  - Gender that's not M/F
- Some tools will identify outliers – be sure you know now they make the determination

Outliers

- How can you tell an outlier from an error?
  - You often can't (not easily)
  - Example: You are looking for fraud cases. The outliers may contain those odd, extreme data points typical of fraud cases
- Don't discard outliers unless you're sure they are really outliers.
  - You may be looking for data points comprising only 1% of your data

Outliers

- Best approaches
  - Work with your domain expert
  - Try to help identify why the data values are extreme
  - Do remove outliers if you think they will negatively impact your analysis
  - Check the source & quality of the raw data

Outliers

- One last word:
  - Many people rely on data warehouses as a source of data
  - Data in warehouses is typically “cleaned”
  - Find out if outliers have already been removed or you may be looking for those 1% fraud cases and the evidence is already gone!

Visualizations

- Some tools provide visualizations for you to help understand the data
- Some are very good – a picture IS worth 1000 words
- One caution: Suppose your tool provides a simple line chart of the data. It may be difficult to see relationships with fields not adjacent to each other – you may need to reorder the data

Transforms

- You may encounter situations where you need to transform your data
  - Home prices (a “normal-looking” distribution but skewed tail on the right due to extremely high home prices (not on left because scaled to zero)
  - Donation amounts (same idea)

Transforms

- May need to transform the data to make neater bins or to scale for visualizations
- One approach is to apply $\log_{10}$ function to numeric values

Log_{10} Transform

- To address skew (house price example)
- $\text{Log}_{10}\ 1 = 0$
- $\text{Log}_{10}\ 10 = 1$
- $\text{Log}_{10}\ 100 = 2$
- $\text{Log}_{10}\ 1000 = 3$

- What is $\text{Log}_{10}\ 0$?

Log_{10} Transform

- What is $\text{Log}_{10}\ 0$? Undefined
- We don’t want 0 so bump all up by 1
- Now, $\text{Log}_{10}\ 0 = 1$, $\text{Log}_{10}\ 1 = 1$, etc…
- What about negative values? Also undefined so take absolute value
- Scales the data making it easier to visualize and handle

Log_{10} Transform

- In the general form:
- $\text{Log}_{10} |X + 1|$
- Add sin to regain the sign on negative side
- $\text{Sin X Log}_{10} |X+1|$
- Many commercial and open source tools will automatically do this for you

Regression Modeling

- Try to fit the data to a line
- Calculate the error
- Sometimes easy to identify extreme values (but be careful – they may not be outliers!)

The Curse of Dimensionality

- Many types of data analysis become significantly harder as the dimensionality of the data increases
- As the dimensionality increases, the data becomes increasingly sparse in the space it occupies

The Curse of Dimensionality

- Some algorithms (namely classification and clustering) tend to be less accurate with high dimensionality data
- Example: Clustering points. In a 2-d plane, the distance calculation is meaningful. What happens if we look at data from 10 dimensions? The distance becomes less meaningful

Pruning the data – dimensionality reduction

- Data with many attributes often difficult to work with
  - Most learning algorithms look for non-linear combinations of fields
  - Can find spurious combinations given a small number of data records and a large number of attributes

Pruning the data

- Also:
  - Many attributes may not matter
  - Many algorithms perform better if we reduce the number of attributes being considered
- How can we prune the attributes?

Pruning

- Remove fields with little or no variability
  - If all values are the same, the attribute does not matter (much)
  - A possible rule may be to remove those less than 5% variance
- Select the N most important

Pruning

- Principal Component Analysis
- We won’t cover in this course, but PCA essentially projects the data from high dimensionality space into a lower dimensionality space

Simple Pruning Techniques

- Look for redundant data
  - Do you need state and zip code?
  - Do you need purchase price and amount of sales tax?
- Look for irrelevant features
  - Student ID numbers, account numbers (unless metadata embedded in structure of account numbers)

Feature Weighting

- You might also find that some features (attributes) are more/less important than others
- You may want to add/reduce the weighting of select features

Data Cleaning Problems

- Error correction and missing information is a challenging problem
- Incorrect or sloppy data cleaning can result in removing a large amount of data, or incorrect data. Neither one is a good thing!

Data Cleaning Problems

- Often times, there is not enough information or application knowledge to know what to do – consult your domain expert
- Data cleaning is an iterative, time consuming
- Arguably the most time consuming, painful and expensive part of data mining

Data Management Policy

- Policy is needed because:
  - After cleaning & removing errors, you don't want to repeat it if the collection changes or when new values are added
  - When you clean the data, you need to document what you did to the data
  - Need a way to audit and measure the resulting data

Other Data Issues

- Data integrity – analysis is only as good as the data. How can we ensure the data is valid and complete?
  - Data quality – "noise" – modification of original values, outliers, missing values, duplicates
  - Cost – Continuing pressure for larger and faster systems. Driven by the quantity of data and the need for results in a limited amount of time

A Quality Issue:

- Sometimes, classes have very unequal frequency
  - attrition prediction: 97% stay, 3% attrite
  - medical diagnosis: 90% healthy, 10% disease
  - eCommerce: 99% don't buy, 1% buy
  - Security: >99.99% of Americans are not terrorists
- Majority class classifier can be 97% correct, but useless

Metadata

- Is there information about the data embedded in the data itself?
- Example: A data value that is not available/valid unless a specific condition or other related data exists

Metadata

- Metadata encodes background knowledge
- Can be used to restrict search space
- You may need to expand metadata attributes into multiple attributes

Questions to ask

- What data is available? Where will it come from?
- Is the data current/relevant?
- Is other data available?
- Is historical data available?
- Who is the domain expert?
- Which attributes are really important?
  - Remove irrelevant & redundant

How do you know if attributes are irrelevant?

- Irrelevant attributes have no influence on the outcome (e.g. Customer ID)
- May slow things down
- May be illegal to use (race, SS #)
- Not always easy (Abbott):
  – A company reported increased sales during certain phases of the Moon. Had nothing to do with the Moon but sure looked like it

Using the Data

- We never use the entire data set to train the algorithm
  - Doing so runs the risk of overfitting the data
  - When the model is built to fit the training set, it loses the ability to generalize to new data

Using the Data

- Typically, the data set is split into a training set and a test set
  - In theory: About 70% of the data for the training; 30% for the test set
  - In practice: Roughly 50/50 for training and test and reuse for validation
- Should also split into a third validation set
  - Use to verify the results or tune the algorithm parameters

Using the Data

- Here is a good example (Abbott):
  - You are preparing to take the GRE, buy a book with answers in the back (Training)
    - Study, take test, study, take test …
  - But do you really know the material? – go out and buy a second book (Testing)
  - Validation is how you do when you take the "real" exam

Sampling

- How do you build your testing, training and validation data sets?
  - Random without replacement
  - Systematic sample
    - Select every Nth instance
    - Don't select first N – order may influence the data

Sampling

- Stratified Random Sample
  - Select equal number of instances from each attribute's values
    - Same number of males/females
    - Same number of fraud/not fraud
  - Why? If looking for a small #, especially with a small N, you are training on biased data

Sampling Example

- You are looking for the 1% of people who give > 100K to a specific charity
- You randomly pick 5000 records.
- How many large donors are you likely to have in your sample?

Sampling Example

- You are looking for the 1% of people who give > 100K to a specific charity
- You randomly pick 5000 records.
- How many large donors are you likely to have in your sample?
- If you randomly pick, you are unlikely to get ANY of them. Your model will look great – on paper, but you won't have the results you want.

Sampling

- For all sampling, think counts, not percentages. Need enough data to safely generalize
- Otherwise, no general rules
  - How complex is the data & problem?
  - How much noise?
  - Are there duplicate patterns?
  - How many attributes?
  - Are there underrepresented data values?

Data Warehouse

- A typical data warehouse contains normalized data
- Data has been cleaned
- May come from many sources
- Be sure you know how the data may have been cleaned

Summary

- The level of cleaning may depend on the model you plan to use
  - Neural nets, regression, nearest neighbor require that you fix missing data and expand categorical data
  - Trees and rules are less work (on the data prep side) but may not always be the best tool

Domain Experts

- Just as with software development (project champion), you need a domain expert
- Help you understand the scope and business issues
- Can also give advice on what to do with dirty or missing data

Summary

- GI GO
- Good data prep is key to producing valid and reliable models

Where can I find data?

- Many government agencies – you need to look around
- IMDB – popular place students go, but files tend to be *very* large, fragmented (because it’s structured for database tables)
- UCI Machine Learning Repository – lots of data here; watch file sizes – some are too small

Where can I find data?

- Kdnuggets.com
- Google
- Gaming sites
- Twitter
- - you need to look around

Credits

- Dean Abbott
- Data Mining: Practical Tools and Techniques. Witten & Frank. Morgan Kaufmann, 2005.